

```
import numpy as np

# Sample image matrix
image = np.array([
    [10, 10, 10],
    [10, 50, 10],
    [10, 10, 10]
])

# Laplacian mask
laplacian = np.array([
    [0, -1, 0],
    [-1, 4, -1],
    [0, -1, 0]
])

# Apply mask
output = np.sum(image * laplacian)

print("Input image:")
print(image)

print("Laplacian output at center:", output)
```

```
Input image:
[[10 10 10]
 [10 50 10]
 [10 10 10]]
Laplacian output at center: 160
```

